

```

my_accessory.c
// format: string; HAP section 9.62; max length 64
homekit_characteristic_t cha_name = HOMEKIT_CHARACTERISTIC_(NAME, "Switch");

homekit_accessory_t *accessories[] = {
    HOMEKIT_ACCESSORY(.id=1, .category=homekit_accessory_category_switch, .services=(homekit_service_t*[]) {
        HOMEKIT_SERVICE(AccessoryInformation, .characteristics=(homekit_characteristic_t*[]) {
            HOMEKIT_CHARACTERISTIC(NAME, "Switch"),
            HOMEKIT_CHARACTERISTIC(MANUFACTURER, "Arduino HomeKit"),
            HOMEKIT_CHARACTERISTIC(SERIAL_NUMBER, "0123456"),
            HOMEKIT_CHARACTERISTIC(MODEL, "ESP8266/ESP32"),
            HOMEKIT_CHARACTERISTIC(FIRMWARE_REVISION, "1.0"),
            HOMEKIT_CHARACTERISTIC(IDENTIFY, my_accessory_identify),
            NULL
        }),
        HOMEKIT_SERVICE(SWITCH, .primary=true, .characteristics=(homekit_characteristic_t*[]) {
            &cha_switch_on,
            &cha_name,
            NULL
        }),
        NULL
    })
};

```

Fig. 10: The highlighted code to be copied and pasted

```

homekit_characteristic_t cha_name = HOMEKIT_CHARACTERISTIC_(NAME, "Switch");

homekit_accessory_t *accessories[] = {
    HOMEKIT_ACCESSORY(.id=1, .category=homekit_accessory_category_switch,
        HOMEKIT_SERVICE(AccessoryInformation, .characteristics=(homekit_characteristic_t*[]) {
            HOMEKIT_CHARACTERISTIC(NAME, "Switch"),
            HOMEKIT_CHARACTERISTIC(MANUFACTURER, "Arduino HomeKit"),
            HOMEKIT_CHARACTERISTIC(SERIAL_NUMBER, "0123456"),
            HOMEKIT_CHARACTERISTIC(MODEL, "ESP8266/ESP32"),
            HOMEKIT_CHARACTERISTIC(FIRMWARE_REVISION, "1.0"),
            HOMEKIT_CHARACTERISTIC(IDENTIFY, my_accessory_identify),
            NULL
        }),
        HOMEKIT_SERVICE(SWITCH, .primary=true, .characteristics=(homekit_characteristic_t*[]) {
            &cha_switch_on,
            &cha_name,
            NULL
        }),
        NULL
    })
};

```

Fig. 11: Changing the id

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HOMEKIT_CHARACTERISTIC(FIRMWARE_REVISION, "1.0"),
HOMEKIT_CHARACTERISTIC(IDENTIFY, my_accessory_identify),
NULL
}),
HOMEKIT_SERVICE(SWITCH, .primary=true, .characteristics=(homekit_characteristic_t*[]) {
    &cha_switch_on,
    &cha_name,
    NULL
}),
NULL
});

```

Fig. 12: Changing the channel switch number

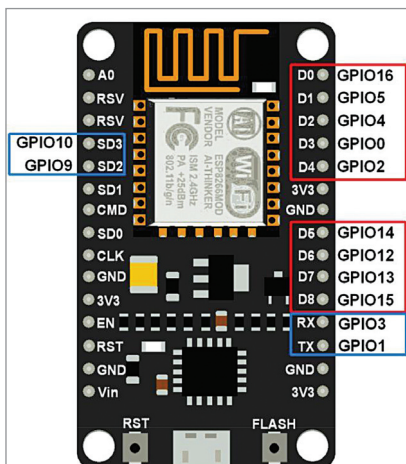


Fig. 13: GPIO pins on NodeMCU board

```

#define PIN_SWITCH 2
#define PIN_SWITCH2 16
#define PIN_SWITCH3 4
#define PIN_SWITCH4 5

```

Fig. 14: GPIO pins defined in the code

```

cha_switch_on.setter = cha_switch_on_setter;
cha_switch_on2.setter = cha_switch_on_setter2;
cha_switch_on3.setter = cha_switch_on_setter3;
cha_switch_on4.setter = cha_switch_on_setter4;

```

Fig. 15: Changing the switch_on and setter in the code

In the highlighted part you will find “&cha_switch_on” line, as shown in Fig. 12. This is default code for channel switch number. If you added a new highlighted part, change it to “&cha_switch_on2”. You need to increase the switch number every time you add a new highlighted part.

Next, you will find “homekit_characteristic_t cha_switch_on = HOMEKIT_CHARACTERISTIC_(ON, false);” in the same code. Here too, you need to add this line as many times as you have added the devices and also add the switch_on number. For example, if you added two devices, the line of code will be as follows:

```

homekit_characteristic_t cha_switch_on = HOMEKIT_CHARACTERISTIC_(ON, false);
homekit_characteristic_t cha_switch_on2 = HOMEKIT_CHARACTERISTIC_(ON, false);

```

5. Now make some changes in Example02_Switch code. Here, you will find the line “extern “C” homekit_characteristic_t cha_switch_on;” You need to add this line every time you add a device and also, as before, you have to change switch_on to switch_on2, switch_on3, etc.

Next, you will find a “#define PIN_SWITCH 2” line of code. Copy this line and add as many times as you have added new devices and also change each pin number. This pin actually defines the pin of NodeMCU board where the device is connected through the relay module.

Note that the pin 2 defined here is not the physical pin D2 of NodeMCU; it is GPIO pin 2. The GPIO pins are marked on NodeMCU board, as